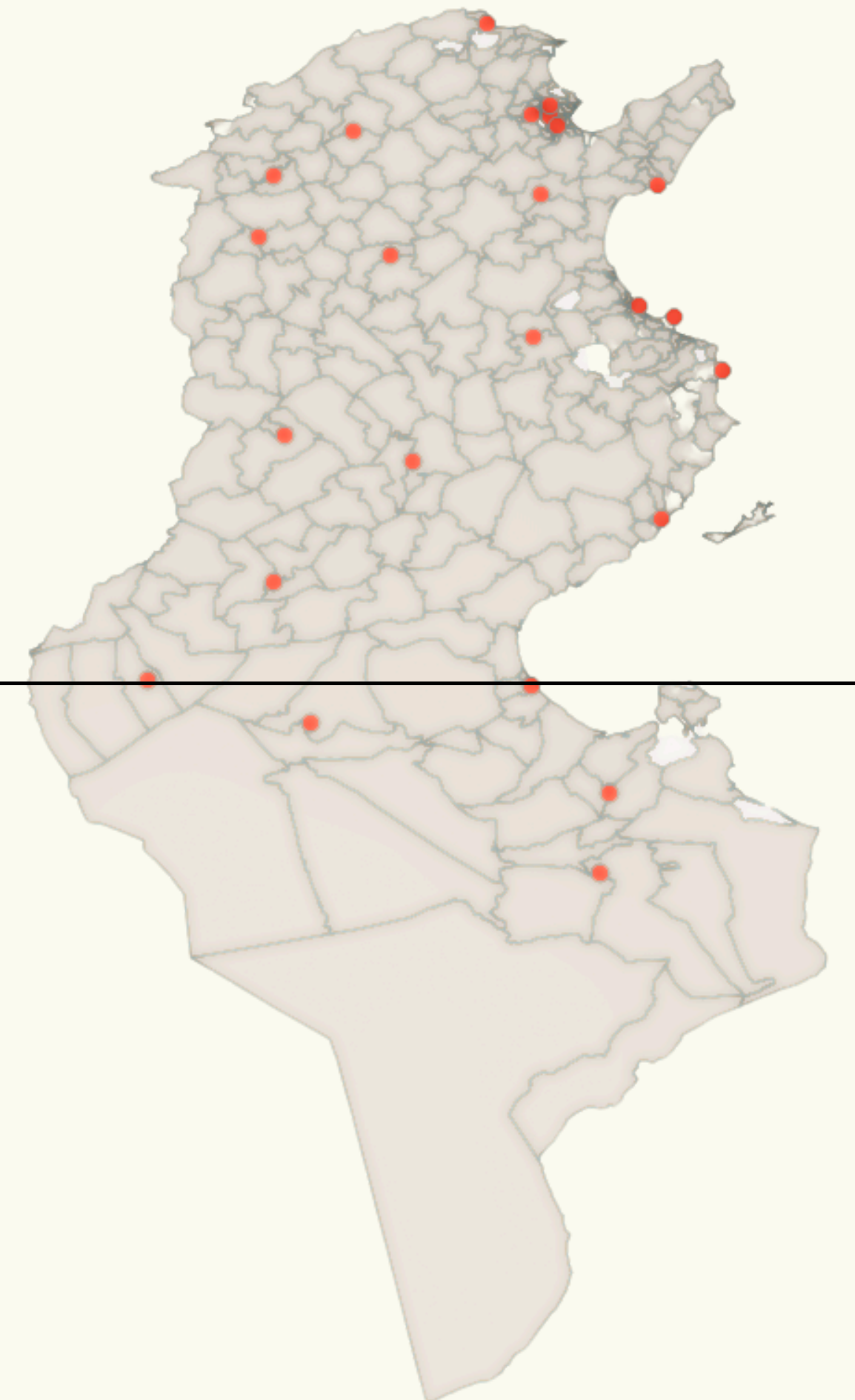


AGRICULTURAL DATA

Harmonization

& Geospatial Metadata Framework

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Data source:

- ONAGRI

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EXECUTIVE SUMMARY :::::

This report documents a metadata-centered methodology for converting raw Tunisian agricultural datasets into standardized GIS-ready outputs.

Our work addresses a recurring problem in Tunisian open data: datasets are often useful statistically, but they are not immediately usable for spatial analysis because administrative names, language, field structures, units and geographic identifiers are inconsistent.

PROJECT OUTCOMES

Outcome type	Description
Operational outcome	Raw tabular datasets are converted into clean datasets and shapefiles.
Governance outcome	A common metadata structure makes datasets traceable, comparable and maintainable.
Analytical outcome	Datasets can be joined to boundaries, mapped, compared across regions and integrated into dashboards.
Risk reduction outcome	Future datasets can be collected using controlled region codes, multilingual dictionaries and validation rules, reducing manual cleaning.

IN-SCOPE DATA TYPES

Data type	Examples
Administrative statistics	Data summarized by governorate, delegation, major region or a local administrative unit.
Facility inventories	Milk centers, storage warehouses, extension cells and other named service points or assets.
Resource and infrastructure datasets	Dams, hydraulic resources, irrigated perimeters and treated wastewater irrigation perimeters.
Production and environmental datasets	Livestock, prices, forest production, forest fires, land vocation and agricultural surface data.
Time series	Annual or monthly observations that require explicit time-period metadata and consistent temporal fields.

METADATA DESCRIPTION FRAMEWORK

Metadata layer	Purpose and required content
A. Identification metadata	Uniquely identifies the dataset and its version. Includes dataset ID, title, category, source organization, collection date, publication date, version, owner/contact and access path.
B. Thematic metadata	Explains the agricultural topic. Includes category, subcategory, statistical concept, population observed, indicator meaning and intended use.
C. Temporal metadata	Explains time coverage. Includes year, month, reporting period, update frequency, time granularity and whether the dataset is cross-sectional or time-series.
D. Geographic metadata	Explains the spatial reference. Includes geographic level, governorate/delegation fields, official region_id, coordinates, CRS, boundary source, matching method and match confidence.
E. Attribute and variable metadata	Defines each column. Includes standard field name, original field name, language, data type, unit, allowed values, missing-value codes, transformation rule and quality notes.
F. Processing and lineage metadata	Documents what happened to the data. Includes raw file name, processing date, normalization dictionary used, join key, transformations applied, validation checks and output shapefile path.

DATASET & THEMATIC METADATA

Dataset Field	Description
dataset_id	Stable unique identifier.
dataset_title	Human-readable title in English/French/Arabic where relevant.
category	High-level folder or theme.
source_name	Name of institution, portal or CRDA source.
source_url_or_catalog_reference	Original catalog link or local reference path.
raw_file_name	Original file exactly as received
data_owner_contact	Institution/person responsible for the dataset.
license_access	Access conditions, license or reuse restrictions.
version	Dataset version and whether it is raw, cleaned, normalized or GIS-ready.
processing_status	Raw, cleaned, standardized, joined, exported, validated or needs review.

Thematic Field	Description
theme	Broad policy/technical domain.
subcategory	More specific dataset family, such as livestock statistics, public irrigated perimeters.
indicator_family	Production, infrastructure, resource, price, area, population, facility count, service network.
measurement_subject	What is being counted or measured: animals, hectares, m3, dinars/kg, etc.
analysis_use	Expected use: map, regional comparison, trend analysis, dashboard indicator.

TEMPORAL & GEOGRAPHIC METADATA

Temporal Field	Description
time_coverage_start / end	Earliest and latest period included.
time_field_original	Original field name, such as Annee, Année, Date, mois.
time_granularity	Annual, monthly, mixed annual/monthly, or single reporting year.
period_type	Observation year, approval date, exploitation year, etc.
date_conversion_rule	How Excel serial dates, text months and mixed date formats are converted.
update_frequency	One-off, annual, monthly or unknown.

Geographic Field	Description
geo_level_original	Spatial level found in the raw data.
geo_level_standard	Standardized GIS level used for output.
region_name_original	Administrative name as written in the file.
region_name_standard_fr / ar	Official French/Arabic name used for joining.
region_id	Stable synthetic or official key.
geometry_source	Boundary file or point coordinate source.
coordinate_fields	Latitude/longitude/GPS columns if present.
crs	Coordinate reference system (e.g., EPSG:4326).
match_method	Exact, dictionary, fuzzy, manual coordinate.
match_confidence	High, medium, low or unresolved.
join_status	Joined, no match, duplicate match, aggregate.

VARIABLE & PROCESSING METADATA

Variable Field	Description
original_field_name	Column name as received, including original language and spacing.
standard_field_name	Clean column name used in outputs.
language	Arabic, French, English, mixed or code-like.
data_type	Text, integer, decimal, date, coordinate, categorical, boolean or note.
unit	ha, m3, million m3, head, litre, dinar/kg, etc.
allowed_values	Controlled values for categories, statuses, etc.
missing_value_rule	How nulls, blanks, hyphens, and free-text missing values are interpreted.
aggregation_rule	Sum, average, count, last value, not aggregatable.
quality_flag	Valid, corrected, converted, suspicious, etc.

Processing Field	Description
processing_script	Name/version of processing step.
processing_date	Date the output was produced.
reference_dictionary	Version of administrative gazetteer or lookup table used.
boundary_file	Boundary layer used for geometry.
transformations	Renaming, translation, unit parsing, numeric conversion, etc.
validation_checks	Checks for missing keys, duplicated regions, invalid coordinates, encoding errors.
output_file	Clean dataset path and final shapefile path.
reviewer_status	Auto-validated, manually validated, partial match, rejected or pending.

METHODOLOGY

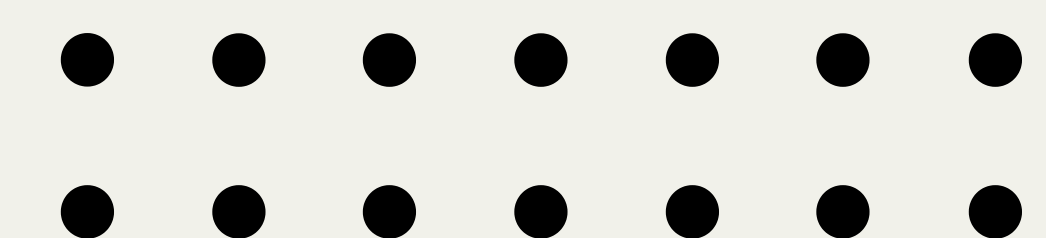
1- All datasets are downloaded by hand from the ONAGRI-related data catalog website (<https://catalog.agridata.tn/en/dataset/>) for each category (Forests, Animal production, Land development, Dams & Irrigated areas, Rainfall, Agricultural products).

2- The first portion of the R code fixes the major inconsistencies between Arabic and French naming conventions in the raw data and normalizes it using a reference dataset "Tunisia.csv" (extracted from Tunisia GIS files, source: <https://gadm.org/>).

A unified **region_id** is added to join tabular data with spatial geometries. The output datasets are stored automatically as **Clean data**.

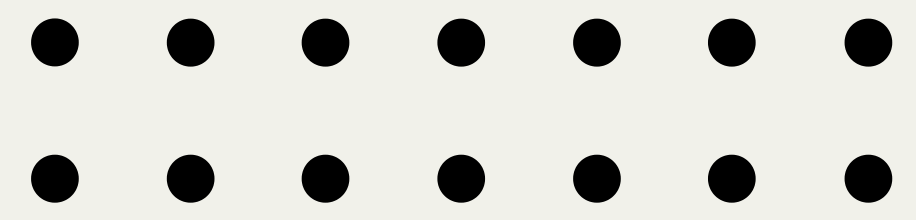
3- The second portion of the R code merges the clean data with the Tunisia_regions.shp boundary file using region_id. This attaches geometry to non-spatial data.

4- The output **shapefiles** are stored to be used directly by any GIS software.



Our work environment:

https://drive.google.com/drive/folders/1hRmI436IKVzJ2eBKvzt8iuPr5_CaFmTp?usp=drive_link



WORKFLOW GUIDE

0– Ensure all datasets are in their proper folders and the file paths in the code are correct.

1– Run "**1– Data processing & normalization.R**". Check for results in the "Clean data" folder.

2– Run "**2– Conversion into GIS data.R**". Check for results in the "GIS Layers" folder.

3– Go to QGIS (or any GIS software) and load the **Tunisia_map.shp** found in the "Tunisia_map" folder. This will act as a reference map.

4– In QGIS, go to **Layer › Add Layer › Add Vector Layer › select any shapefile from the "GIS Layers" folder**. The points will be placed on the map.

You can then **click on the layer › Show Labels** to fully display the data.

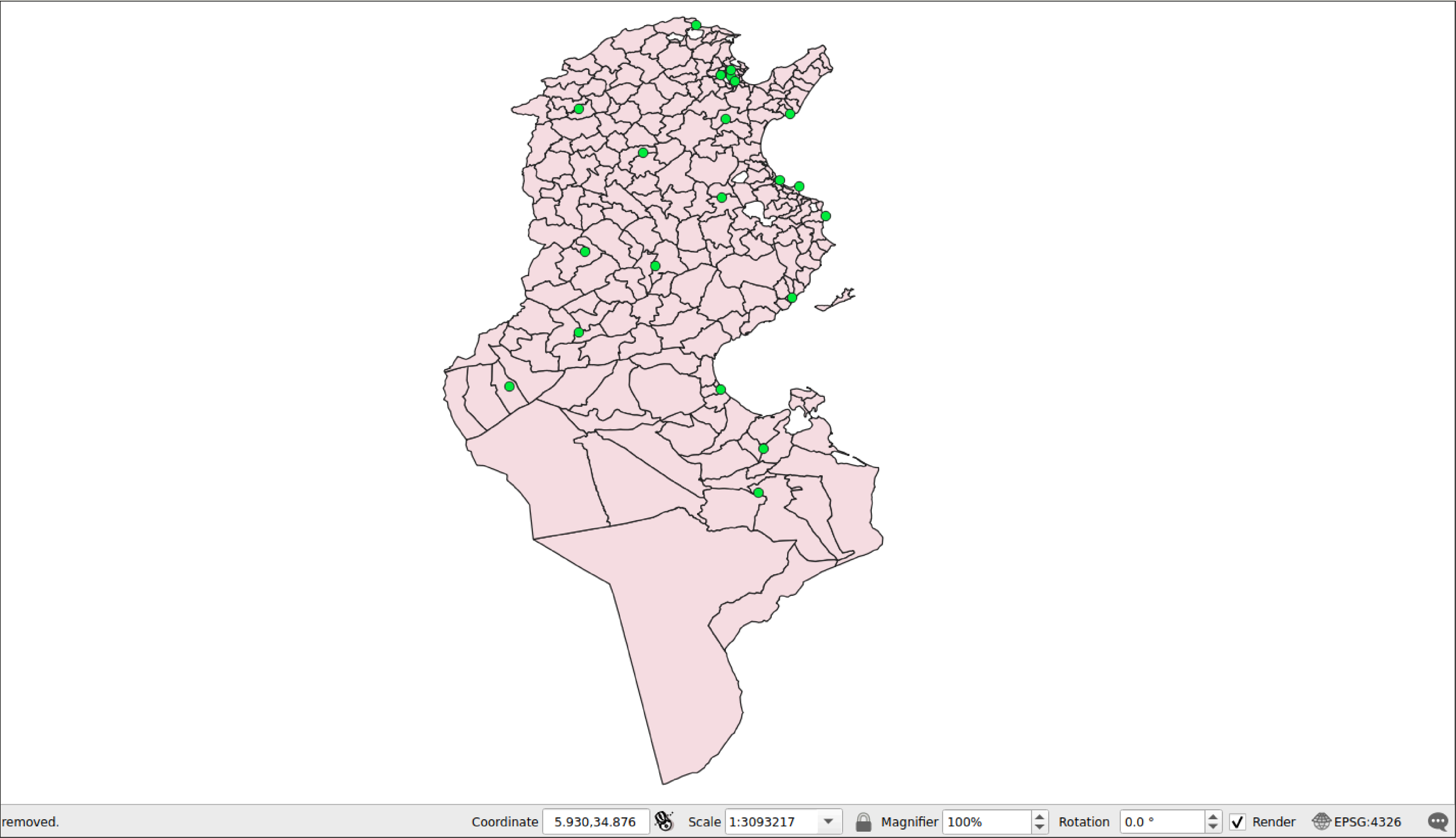
DELIVERABLES & OUTPUTS

Raw-to-clean Deliverable	Purpose
Raw data folder	Original downloaded datasets preserved without modification.
Clean data folder	Standardized tabular outputs after naming, typing, unit and region normalization.
GIS Layers folder	Shapefiles generated from clean datasets after geometry attachment.
Tunisia_map folder	Reference administrative map used for visual context in QGIS.
Reference dictionary	Tunisia.csv or equivalent table containing official administrative names and region_id.
Metadata inventory	Central record describing each raw, clean and spatial output.

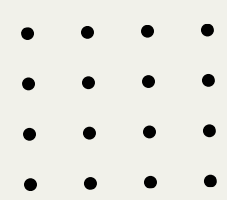
Output component	Purpose
.shp/.shx/.dbf/.prj	Standard shapefile components.
_metadata.xlsx or .csv	Dataset-level and field-level metadata.
_quality_report.txt or .html	Validation results, unresolved joins and warnings.
_lineage.json	Machine-readable transformation and source lineage.
_readme.txt	Human-readable explanation of data source, limitations and intended use.

FINAL OUTPUT EXAMPLE:

This demonstrates how a non-spatial agricultural table becomes a spatial layer after metadata and region matching.



From the dataset: Distribution of forest land areas by governorate.



MAIN CHALLENGES

Challenge	Why it matters
Inconsistent region names	Administrative names appeared in Arabic, French, transliteration and variant spelling. This makes direct joins unreliable.
No stable region identifier	Many datasets did not include official codes, forcing creation of a synthetic region_id from a reference dataset.
Mixed administrative levels	Some files use governorates, some delegations, some facilities, some major regions and some totals. These cannot be mapped with one universal method.
Geography stored in filenames	Some datasets do not include governorate fields but the file title contains the location. This context is easy to lose.
Coordinates are inconsistent or absent	Some files include latitude/longitude; others contain GPS strings; many contain no coordinates.
Language and encoding issues	Arabic and French fields can create encoding, sorting, matching and normalization problems.
Variable naming is not standardized	Fields such as Annee, Année, annee, Date or year-like columns need standard mapping.
Unit ambiguity	Units appear inside field names, values or not at all. This creates risk when comparing datasets.
Totals mixed with detail rows	Rows such as Total مكة can distort maps if treated as normal delegation records.
Missing values use multiple symbols	Blank cells, hyphens, underscores, zero values and text notes may represent missingness differently.

PREVENTION & FIELD CHECKLIST

Problem	Preventive control
Region-name inconsistency	Official administrative gazetteer, bilingual dictionary, dropdowns at data entry and automated region_id matching.
Mixed language	Mandatory bilingual name fields, approved transliteration rules and UTF-8 encoding checks.
Missing geography	Mandatory geo level and region id fields; coordinate requirements for facility datasets.
Unit inconsistency	Approved unit dictionary and data-entry templates that separate value from unit.
Temporal inconsistency	Controlled year/month fields, ISO dates and standardized reporting-period definitions.
Duplicate records	Composite uniqueness rules such as dataset_id + year + month + region_id + indicator.
Ambiguous missing values	Single approved missing-value convention and machine-readable nulls instead of hyphens or underscores.

Metadata field	Priority	Description
dataset_id	Required	Unique dataset key.
dataset_title	Required	Official title.
category	Required	Theme/folder.
source_institution	Required	Data owner or publisher.
raw_file_name	Required	Original filename.
time_coverage_start	Required if temporal	Start year/month/date.
geo_level	Required	Governorate, delegation, point, major region, national.
region_id_field	Required for spatial join	Column used for geometry join.
coordinate_fields	Required for point data	Latitude/longitude or GPS fields.
crs	Required for coordinates	Coordinate reference system.
variable_dictionary	Required	Field-level metadata exists.
unit_dictionary	Required for indicators	Unit definitions.
processing_status	Required	Raw/cleaned/joined/exported/validated.
output_layer_name	Required for GIS	Final shapefile name.

VARIABLE DICTIONARY TEMPLATE

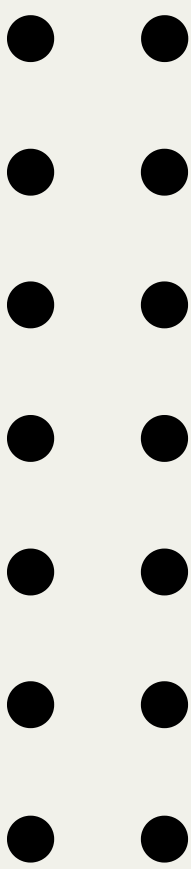
Original field	Standard field	Language	Type	Unit	Allowed values	Rule	Quality note
Délégation	delegation_name_fr	French	text	n/a	official delegation list	trim and match to dictionary	valid after matching
المعتمدية	delegation_name_ar	Arabic	text	n/a	official delegation list	match Arabic name to region_id	requires UTF-8 check
Superficie irriguée (ha)	irrigated_area_ha	French	decimal	ha	positive decimal	decimal comma converted to dot	against irrigable area
longitude	longitude	French	decimal	degrees	-180 to 180	numeric	coordinate conversion range check

SAMPLE DATASET INVENTORY

File	Sheet	Rows	Cols	Language	Geo pattern	Example fields
Milk collection centers - Ariana Governorate.xlsx	Feuill	6	8	French/English	coordinates; delegation	Name, Delegation, Capacité litre, Date_Initiale_d'activité, Date Agrément Sanitaire,...
Milk collection centers.xlsx	Feuil2	116	4	Arabic/French	delegation; governorate	الولاية, المعتمدية . عدد مراكز تجميع الحليب
List of territorial extension cells Governorate of Tataouine.xlsx	Feuill	9	3	French/English	coordinates	CTV, Téléphone, GPS
Livestock statistics Sidi Bouzid-2021.xlsx	Feuill	15	14	Arabic/French	delegation	المعتمدية, عدد الابقار مؤصلة, عدد الابقار اخرى العدد الجملي للابقار, عدد القطيع (انثى منتجة) : أغنام (ألف)
Evolution of the sheep herd - Governorate of Sousse.xlsx	ovins	129	4	French/English	delegation	Annee, Délégation_Fr, Délégation_Ar, Nombre_femelle_productive
Quantities of water produced by SECADENORD in million m3.xlsx	Feuill	100	5	French/English	coordinates	Annee, Barrage, Latitude, Longitude, Quantite d'eau_produite_par_barrage_million_m3

CONCLUSION

The most important conclusion is that metadata must become a first-class deliverable. Without metadata, the shapefiles are technically usable but difficult to audit, reproduce, update or compare. A strong metadata framework should describe each dataset, each variable, each geographic reference, every transformation rule, and the quality status of the output.



THANK YOU.

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